Factorial using Iterative Method

```
import time

n = int(input("Enter a number: "))

start = time.time()

factorial = 1

for i in range(1, n + 1):
    factorial = factorial * i

end = time.time()

print("Factorial =", factorial)
print("Execution Time =", end - start, "seconds")

print("Time Complexity: O(n)")
print("Space Complexity: O(1)")
print("Best Case: O(n)")
print("Average Case: O(n)")
print("Worst Case: O(n)")
```

```
Enter a number: 45
Factorial = 119622220865480194561963161495657715064383733760000000000
Execution Time = 0.00018167495727539062 seconds
Time Complexity: O(n)
Space Complexity: O(1)
Best Case: O(n)
Average Case: O(n)
Worst Case: O(n)
```

Factorial using Recursive Method

```
import time

def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)



n = int(input("Enter a number: "))

start = time.time()

result = factorial(n)
```

```
end = time.time()

print("Factorial =", result)
print("Execution Time =", end - start, "seconds")

print("Time Complexity: O(n)")
print("Space Complexity: O(n)")
print("Best Case: O(n)")
print("Average Case: O(n)")
print("Worst Case: O(n)")
```

```
Enter a number: 23
Factorial = 25852016738884976640000
Execution Time = 0.00017952919006347656 seconds
Time Complexity: O(n)
Space Complexity: O(n)
Best Case: O(n)
Average Case: O(n)
Worst Case: O(n)
```

BOTH METHODS IN ONE PROGRAM

```
import time

# Iterative method
def factorial_iterative(n):
    result = 1

    for i in range(1, n + 1):
        result = result * i

    return result


# Recursive method
def factorial_recursive(n):
    if n == 0 or n == 1:
        return 1

    return n * factorial_recursive(n - 1)


n = int(input("Enter a number: "))

# Iterative
start = time.time()
result1 = factorial_iterative(n)
end = time.time()

print("\n--- Iterative Method ---")
```

```
print("Factorial =", result1)
print("Execution Time =", end - start, "seconds")
print("Time Complexity = O(n)")
print("Space Complexity = O(1)")


# Recursive
start = time.time()
result2 = factorial_recursive(n)
end = time.time()

print("\n--- Recursive Method ---")
print("Factorial =", result2)
print("Execution Time =", end - start, "seconds")
print("Time Complexity = O(n)")
print("Space Complexity = O(n)")
```

```
Enter a number: 23

--- Iterative Method ---
Factorial = 25852016738884976640000
Execution Time = 0.00025463104248046875 seconds
Time Complexity = O(n)
Space Complexity = O(1)

--- Recursive Method ---
Factorial = 25852016738884976640000
Execution Time = 6.127357482910156e-05 seconds
Time Complexity = O(n)
Space Complexity = O(n)
```